Notes for guidance on final year projects

Final year projects provided by the Department of Informatics can vary dramatically in their subject matter from software-based projects to media creation projects. Thus, guidance from one project to the next will necessarily be different. However, there are many aspects that all projects must share to achieve a good mark.

MANAGEMENT AND PLANNING - When involved with any project it is important to manage your time and resources efficiently. One of the tasks in the first weeks of your project should be to produce a timetable for your project. This will usually be in the form of a "Pert" or "Gantt" chart and will show the amount of time spent on each activity. It should also include dates when important milestones should take place e.g. completion of background research stage, start of prototyping, completion of interim report (week 7) etc.

I would recommend putting a calendar event in your diary so that you weekly know to work on your project.
Note: the interim report contains the • *Introduction* • *Professional Issues* • *Background Research* • *Requirements Specification* • *Planning chapter*

Although it is hard at the start of the project to estimate how long particular stages of the project will take it is important that you do produce a timetable to guide you.

PHYSICAL RESOURCES - Early on in your project you will be making a decision on the physical resources that you need for your project e.g. which computer. Whether you choose to use the Department's machines or your own PC you must take the utmost care to make frequent backups (I suggest using OneDrive or GitHub) of your application and dissertation.

I would advise you to work on university computers because this gives you access to some technical support that might be useful.
It would help if you got into the habit of saving work every 5 minutes while working on your projects.
It would help if you gave files appropriate names – this does make it easier

Loss of work due to computer problems and your failure to make proper back-ups is not eligible as exceptional circumstances.

PROJECT STRUCTURE - your project should contain the following activities.

Background literature review - this is a review of a reasonable portion of the available literature that is relevant to your chosen project. This literature might be in the form of books, conference and journal papers, magazine and newspaper articles, web pages etc. This literature review will enable you to create the academic justification for your project and project choices.

Your literature review may cover many areas, but should ideally include background for your project idea and a justification of the software and process you plan to use.

Below are some examples of the areas you would need to review for particular projects:

<u>If your goal is to develop an application over the web you would need to have a review of the following:</u> • Similar web-based applications – pick two or three to review and discuss their functionality, advantages and disadvantages. • A review of some appropriate methods for implementing the application on the web e.g. languages, libraries, development tools etc. This should be accompanied by a justification of the specific development stack that you have chosen for your project. • A review of relevant methodologies that are applicable to the specification, design, implementation etc. of your web-based application along with a justification for your choice of methodology.
<u>If your goal is to develop a 3D game, you would need to have a review of the following:</u> • A critique of a few similar games that will help you decide what functionality/gameplay to put into your game. Pick two or three to review and discuss their functionality, advantages and any limitations – you can include game review information in this critique. • How games of the type/genre you intend to implement are produced at the moment. e.g. a review of development platforms/languages (i.e. Unity, UnReal, Java, C++, etc.). This should be accompanied by a justification of the specific development stack that you have chosen for your project. • A review of relevant methodologies that are applicable to the specification, design, implementation etc. of your game along with a justification for your choice of methodology.
<u>If your project is a video using complex visual effects techniques, you would need to have a review of the following:</u> • A review of similar effects shots.

• Different methods of implementing visual effects e.g. methods of combining 3D animation with live action along with justification of your software choice.

• The possible methodologies used to design visual effects and videos containing effects along with a justification for your choice of methodology.

Production - once you have chosen the most appropriate methodology, you can then use it to produce your project. This involves the whole production cycle from feasibility and specification to design, implementation, testing and evaluation.

Testing - this should follow a thorough plan to make sure that your application contains no programming or conceptual errors and that it is functionally correct. i.e. parts of the application perform as expected, all errors and invalid inputs are caught. The strategy for your tests should be linked to your requirements.

Ethical review:

You will need to evidence that you are working in an ethical way and reflect on this in your report. You must make sure all ethical clearance is secured before doing any user testing.

Evaluation - Evaluation of the project normally requires showing it to the prospective end users/viewers and gaining some feedback on the project's suitability, ease of use, functionality etc. Coming at the end of testing (when time is often short) it is easy to miss this component, but it is extremely important for real products and for obtaining top marks. If you plan to use a user evaluation then you will need to get ethical approval for your evaluation – see information here. This needs to go in the appendix of your interim report.

You should also produce an evaluation of the project as a whole for your dissertation (this would normally go in your conclusion). This will include a review in retrospect of how appropriate your actual system and methodology choices were.

DISSERTATION WRITING - it is suggested that this activity takes part to some extent throughout the project. e.g. you can write up the results (possibly just in draft) of your literature survey once it is completed and whilst you are designing the project. It is always a good idea (but hard in practise) to keep detailed notes on what you have done, this will help tremendously when you start to document your project in the dissertation.

Last year, some students lost marks for relying too heavily on tools like ChatGPT when writing their reports. It's important that your work reflects your own understanding so remember that we want to hear about your project from your own words. This is not something that generative AI models do very well.

If you're unsure or would like support, please don't hesitate to reach out. You're always welcome to submit drafts and discuss your report with me at any stage of the process.

Below are some links to support available at the university.

Guide to referencing
Link to guide on different referencing styles.
Harvard referencing style.

Help with language & writing skills
You can find English language support here
You can find essay writing support here

If you know you have a specific learning difficulty such as dyslexia, or think you may have, please contact Disability Support who have advisors with expertise in these areas. The Disability Support pages also have information about other support services relating to studying and assessment.

Try to keep a research journal – I use OneNote for this, but you can use voice recording on your phone as well.

Your dissertation should generally have the following chapters:

- Abstract – a few paragraphs concisely describing your project, i.e. what you did and what the results were.
- Introduction – general introduction to the area of your project
- Background Research/Literature Review/Requirements Analysis – depending on the project this is likely to be either a review of existing systems/web-sites/video effects or coverage of the research literature that is the basis of your work (or possibly a combination of both). Along with a justification of the software you have chosen.
- Professional Considerations – A short review of how the relevant BCS criteria apply to your project and a discussion of the ethics surrounding your usability evaluation.
- Requirements – a list of functional and non-functional requirements for your project
- Design – an overview of the code structure and interface design for your project or a storyboard or similar for an effects project.
- Implementation – this is the core chapter and should describe the project components that you have produced. You can refer to code in this chapter (if you have produced code for your project), but it should not be a line-by-line description of your code and should be a more high-level description of the functionality which gets across the complexity of what you have produced.
- Testing – description of how you have made sure that your work is as bug free as possible.
- Evaluation – preferably a user evaluation of the “usability” of your system.

- Conclusion – should cover 3 components:

1. How successful your project was, i.e. did it fulfil your requirements and meet the project goals

2. Were there any key issues i.e. how might you do things differently if you could complete the project from scratch again

3. Future work – what would you do if you had more time?

References – these should use a recognised style and be cited from the appropriate place in the text - http://www.sussex.ac.uk/skillshub/?id=251

Appendices – any supporting material which does not fit in the main text, e.g. additional design documents, user evaluation questionnaires etc.

Remember: Your project is a substantial part of your Degree, manage it well and you are likely to get a good mark, but above all this is your chance to work in an area in which you are interested and enjoy, make the most of it.

We will schedule regular meetings to support your project but you can email me if you have questions.

Good Luck.